

ECON 131 — Topics, Priorities, and Workload Plan

Topics, Priorities, and Workload Plan

1. Course information

- **Course:** Economics 131 — Public Economics
- **Instructor:** Danny Yagan
- **Textbook (recommended, not required):** Jonathan Gruber, *Public Finance and Public Policy* (any edition). Lectures loosely follow Gruber.
- **Final exam: Fri May 15, 2026 — 3:00–6:00 PM PT (180 minutes)**
 - In-person, written, on paper.
 - **Closed everything: NO notes, NO cheat sheet, NO calculator, NO electronic device.** This is the only exam of your four where memory load is the binding constraint.
 - Cumulative across the full semester (including material tested on Midterms 1 and 2).
 - Final = **40% of grade**. Lower midterm score replaced by final if it improves your grade → final is high-leverage.
 - Curve: ~35% A, 45% B, 15% C, 5% D/F (per syllabus, adjusted by performance).
- **Core idea (one sentence):** Build a toolkit for evaluating government intervention — tax incidence, optimal taxation, externalities, public goods, social insurance — using a small set of empirical tools (DiD, RD, event studies) and a small set of theoretical concepts (elasticity, marginal social benefit/cost, deadweight loss).
- **Critical Yagan rule (per syllabus, paraphrased to avoid quoting):** Know definitions and formula *definitions* (poverty rate, debt evolution equation, elasticities). Do **not** memorize formulas you derive in lecture (e.g., the optimal top tax rate formula). DO know the *lessons* those formulas communicate (e.g., "high taxable-income elasticity → low optimal top rate").

2. Topics — every lecture this semester

Lecture filenames embed the topic and Gruber chapter. There are 19 numbered lectures (01–19); Lecture 13 is missing in the folder (likely a holiday or pre-MT2 review day). Several lectures have companion `_liveanalysisinclass*.pdf` files showing in-class Stata work.

Block A — Foundations (Lec 01–04)

#	Lecture	Gruber ch	Topic / main idea
01	Intro	1	Why government intervenes, course overview, motivation.
02	Theory tools (reference material only)	2	Microeconomic theory toolkit: utility maximization, indifference curves, elasticity. Reference doc, not a full lecture.
03	Empirical tools	3	DiD, RD (regression discontinuity), event studies, identification assumptions. Includes live Stata analysis.
04	Budget and debt analysis	4	Government budget constraint, debt evolution equation, primary deficit, debt-to-GDP dynamics, fiscal sustainability.

Block B — Taxation & Redistribution (Lec 05–10) — the meaty middle

#	Lecture	Gruber ch	Topic / main idea
05	Income distribution	17, 18	Inequality measurement (Gini, top-share), poverty rate definition,

#	Lecture	Gruber ch	Topic / main idea
			redistribution motivations.
06	Tax incidence	19	Statutory vs economic incidence, burden falls on more inelastic side, sales tax / payroll tax incidence diagrams.
07	Optimal labor income tax	20, 21, 2024 update	Mirrlees / Saez-style optimal taxation, top tax rate intuition, equity-efficiency trade-off, ETI (elasticity of taxable income).
08	Labor supply and taxable income	—	Intensive vs extensive margin, EITC, behavioral responses to taxes. 08B Event Study is a methods supplement.
09	Business tax	24 (2026)	Corporate tax, marginal effective tax rate on investment, tax incidence in capital markets.
10	Taxes and savings	—	Life-cycle model, capital income taxation, intertemporal substitution, why a labor tax $\neq$ consumption tax.

Block C — Role of the Government (Lec 11–19)

#	Lecture	Gruber ch	Topic / main idea
11	Externalities	5–6	Pigouvian tax, Coase theorem intuition, deadweight loss from external costs, cap-and-trade vs taxation.
12	Public goods	7 (2026)	Non-rival + non-excludable, Samuelson condition ($\sum MRS = MRT$), free-rider problem, Lindahl pricing.
13	(missing in folder)	—	Likely review/holiday.
14	Local public goods	10 (2026)	Tiebout sorting, fiscal federalism, "voting with your feet," local vs federal provision.
15	Education	11	Human capital theory, signaling vs human capital, public provision rationale.
16	Social insurance	12	Adverse selection, moral hazard, Akerlof lemons, why private insurance markets fail.
17	UI / DI / WC	14	Unemployment insurance, Disability insurance, Workers' compensation. Optimal UI replacement rate (Baily-Chetty intuition).
18	Health insurance	15, 16	Adverse selection in health insurance, moral hazard in healthcare consumption, ACA mechanics, RAND HIE.
19	Social Security	13	Pay-as-you-go vs fully funded, replacement rate, retirement decision, demographic risk.

What you also have (high-signal supplements)

- Section Worksheets/Final Exam Review (From GSI) / — likely the single highest-signal practice doc on the final's actual scope and style. **Read this first.**
- 12 section worksheets (Sections 1–12, with Section 11 marked "skipping" in your folder).
- 3 problem sets (manageable per BER review).
- Past exams: `Final_fall2019.pdf`, `Final_fall2020.pdf` + `Midterm2024_Final.pdf` + solutions.
- Midterm 1 and Midterm 2 prep folders with exam papers.

3. Priority list

ECON 131's final is **conceptual + diagrammatic + algebra-light** (because no calculator). Yagan's exam style is: clear definitions, identify the right framework, show the diagram or the small derivation, state the lesson. Tiers below reflect that.

TIER 1 — HIGH (~50% of ECON 131 time)

These are the topics that recur on every Yagan-format final. They are the spine of the course.

1. **Tax incidence (Lec 06)**. Burden allocation between buyers and sellers based on elasticity ratio, statutory vs economic incidence, payroll tax case. Diagrams + 1-line elasticity argument. *Exam likelihood: ~99%.*
2. **Optimal labor income tax (Lec 07)**. Top tax rate intuition (high ETI → low optimal rate), equity-efficiency trade-off, behavioral response. Yagan's signature topic — concept-only, no formula memorization. *Exam likelihood: ~95%.*
3. **Externalities + Pigouvian taxes (Lec 11)**. Diagrams of marginal social vs private cost, optimal Pigouvian tax = marginal external damage, Coase intuition, cap-and-trade equivalence. *Exam likelihood: ~95%.*
4. **Public goods (Lec 12)**. Samuelson condition ($\sum MRS = MRT$), free-rider problem, why markets undersupply, Lindahl pricing. *Exam likelihood: ~85%.*
5. **Social insurance basics (Lec 16) + UI in particular (Lec 17)**. Adverse selection vs moral hazard distinction, optimal UI replacement rate intuition (Baily-Chetty: replacement rate balances consumption smoothing vs moral hazard). *Exam likelihood: ~90%.*

TIER 2 — MEDIUM (~35% of time)

Likely on the exam, but as smaller items or in combination with Tier 1 questions.

6. **Capital / business tax (Lec 09–10)**. Life-cycle model intuition, why labor tax ≠ consumption tax, marginal effective tax rate, corporate tax incidence. *Exam likelihood: ~70%.*
7. **Government budget constraint + debt dynamics (Lec 04)**. Debt evolution equation $D_{t+1} = (1+r)D_t + G_t - T_t$, debt/GDP dynamics, r vs g comparison. **You MUST know this equation by memory** (per syllabus). *Exam likelihood: ~70%.*
8. **Health insurance (Lec 18)**. Adverse selection in health markets, moral hazard, ACA design intuition, RAND HIE findings. *Exam likelihood: ~65%.*
9. **Social Security (Lec 19)**. PAYGO vs fully funded, replacement rate, retirement age incentives, demographic risk. *Exam likelihood: ~60%.*
10. **Empirical tools (Lec 03 + 08B)**. DiD assumption (parallel trends), RD assumption (no manipulation), event study reading. Often a "interpret this graph" question worth 5–10 points. *Exam likelihood: ~70% as graph-interpretation.*
11. **Income distribution (Lec 05)**. Poverty rate definition (you must know it cold), Gini, top-share measurement. *Exam likelihood: ~60% as definitional question.*

TIER 3 — LOW (~15% of time, mostly skim)

12. **Education (Lec 15)**. Human capital vs signaling debate, public provision rationale. Often 1 small question. *Exam likelihood: ~45%.*
13. **Local public goods + Tiebout (Lec 14)**. Fiscal federalism intuition. *Exam likelihood: ~35%.*
14. **Disability + Workers' Comp specifics (Lec 17 details)**. Specific program details beyond UI. *Exam likelihood: ~30%.*
15. **Theory tools (Lec 02 reference material)**. Pure micro toolkit — should already be cold from Econ 100A. *Exam likelihood: ~20%.*
16. **Intro (Lec 01)**. Skip. *Exam likelihood: ~10%.*

Definitions you MUST know cold (Yagan's explicit syllabus rule)

17. **U.S. poverty rate** — official threshold-based definition by family size.

18. **Government debt evolution equation** — $D_{t+1} = (1+r) \cdot D_t + G_t - T_t$, and the debt-to-GDP dynamics version.
19. **Elasticity definitions** — price elasticity of demand/supply, taxable income elasticity (ETI), labor supply elasticities (intensive and extensive margin).
20. **Tax incidence rule** — burden falls on the more inelastic side.
21. **Deadweight loss** — Harberger triangle, area = $\frac{1}{2} \times \text{tax} \times \Delta Q$.
22. **Samuelson condition** — $\sum \text{MRS} = \text{MRT}$ for efficient public good provision.
23. **Pigouvian tax rule** — set tax = marginal external cost.
24. **Adverse selection vs moral hazard** — adverse selection = pre-contract info asymmetry; moral hazard = post-contract behavioral response.
25. **Optimal UI replacement rate (Baily-Chetty)** — concept only: replacement rate balances consumption smoothing benefit vs moral hazard cost. No formula needed.

4. Summary, workload anticipation, and time-allocation advice

Concise summary

ECON 131 is the **memory-bound** exam of your four — closed everything, 3 hours, cumulative across 19 lectures. The good news: Yagan's exam style is conceptually clean and rewards definition-fluency over algebraic complexity. Your job is to make ~25 definitions and ~8 diagrams stick by Friday May 15. The structural advantage you have is the May 13–14 buffer (16 hours) immediately after your CS finals — that's the actual review window for this course.

Anticipated review workload (honest)

- Total hours for comfortable A-level: **20–24 hours**.
- Master plan allocates **5 hr in 5-day window + 16 hr May 13–14 buffer + ~1 hr May 15 morning ≈ 22 hr**. This is well-calibrated.
- Memorization-heavy material benefits from spaced repetition. The 5 hr in the early window + 16 hr in the late window is actually better than 22 hr all crammed in May 13–14.

Hour-by-hour suggested split within those ~22 hours

During the 5-day window (5 hours total — foundation pass, no cramming yet):

Activity	Hours	Day in master plan
Read Final Exam Review (From GSI) worksheet — top of your list	1.5	Day 4 evening
Tax incidence (Lec 06) — definitions + 1 diagram from scratch	1	Day 4 evening
Optimal labor tax (Lec 07) — high-ETI-low-rate intuition + 2024 update slide	1	Day 4 evening
Externalities + public goods (Lec 11–12) — Samuelson condition, Pigou rule	1.5	Day 5 afternoon + evening

May 13 (Wed, 8 hours — Block 1: Tax & Redistribution):

Activity	Hours
----------	-------

Activity	Hours
Re-read Lec 04 (debt dynamics) — derive debt evolution equation by hand	1
Re-read Lec 05 (poverty + inequality) — memorize poverty rate definition	0.5
Re-read Lec 06–08 (tax incidence + optimal tax + labor supply) — diagrams cold	2
Re-read Lec 09–10 (business + capital tax) — life-cycle model intuition	1.5
Take Final_fall2019 timed (3 hr) — review every wrong answer	3

May 14 (Thu, 8 hours — Block 2: Government Role + drilling):

Activity	Hours
Re-read Lec 11–12 (externalities + public goods) — diagrams + Samuelson	1
Re-read Lec 14–15 (local public goods + education)	1
Re-read Lec 16–19 (social insurance + UI/DI + health + social security) — adverse selection vs moral hazard cold	2
Take Final_fall2020 timed (3 hr) — review every wrong answer	3
Final pass: rewrite the 9 mandatory definitions on a single sheet of paper from memory	1

May 15 morning (~1 hour, light only — NO new material):

- Walk through definitions list one final time on paper.
- Skim Final Exam Review (GSI) one more time.
- Stop studying ~2 hours before exam. Rest. Eat.

Time-allocation advice — adjustments to the master plan

26. **The master plan's allocation is already correct in shape.** The 5/16/1 hour split across the early window / buffer days / exam morning is what closed-everything memorization actually requires.
27. **Take BOTH past finals timed.** They're closed-everything, 3-hour exams — the only realistic calibration tool. Doing them on consecutive days mimics the actual exam fatigue better than spacing them.
28. **The 9 mandatory definitions are your floor.** If you can't write all 9 from memory by Wed evening, push other things to the side until you can. A wrong definition on the final loses easy points; getting them all 100% cold buys you ~10–15 points before the test even starts.
29. **Use blue-book practice paper.** Yagan's exam is paper-based. Practicing on paper (not screen) primes you for the format — handwriting speed actually matters for a 3-hour written exam.
30. **Don't try to memorize Lec 02 reference material.** That's micro theory you've already absorbed. If you've taken Econ 100A, treat Lec 02 as "if I see a confusing word, I look here" — don't pre-load.
31. **If after Final_fall2019 timed you're below 60%,** the diagnosis is one of: (a) definitions weren't cold, or (b) you're wasting time on derivations Yagan said not to memorize. Force-correct by spending May 14 morning ONLY on the 9 definitions + the 8 priority diagrams, even if it means cutting the second timed exam to 90 min.

What NOT to do (anti-advice)

- Don't memorize the algebraic optimal-top-tax formula. Yagan explicitly says don't. Do know "high ETI → low optimal rate."
- Don't review Lec 02 (theory tools) unless something specific in another lecture confuses you.
- Don't write a cheat sheet — you can't bring one. Time spent writing one is wasted; time spent rewriting definitions from memory IS the review.
- Don't try to recreate the full course from the Gruber textbook. The lecture slides + your *Final Exam Review* (From GSI) are sufficient and Yagan-specific.
- Don't underweight definitions because they feel "too easy." Yagan's exams reliably allocate ~20% of points to definitional questions, and those are the easiest points on the test.
- Don't pull an all-nighter on May 14. Sleep is part of memorization.